Model Collapse is a Singular Complexity Trajectory

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Abstract

Recursive training on synthetic data can drive generative models toward collapse, yet most explanations treat collapse as a statistical phenomenon: tail loss, variance shrinkage, or distributional drift. We develop a singular-geometric view of collapse. In a labeled Gaussian mixture abstraction of recursive self-training, we show that mode extinction induces a monotone trajectory in the real log-canonical threshold: as recursive resampling eliminates active modes, the overcomplete mixture moves through increasingly singular strata, with strictly decreasing singular complexity and growing Fisher-information nullity. We further show that this loss of identifiability begins before exact extinction: as a component's weight becomes small, Fisher curvature in its associated parameter directions shrinks proportionally, so exact singularity is the endpoint of a continuous near-nullity trajectory. Thus collapse corresponds not only to reduced distributional diversity, but also to a progressive loss of locally identifiable parameter directions. This theory separates the singular geometry of collapse from the multinomial/Wright–Fisher absorption dynamics. We also prove that, in the mixture abstraction, injecting enough real data prevents component extinction and keeps the singular-complexity trajectory from descending over a fixed horizon with high probability. We then use the theory to motivate loss-landscape diagnostics for neural recursive training, including Hessian spectra, Gauss–Newton effective rank, decoder-block curvature, feature-rank concentration, and local learning coefficient estimates. Across Gaussian mixtures, VaDE, standard VAEs, and a compact DDPM, we find that synthetic-only recursion exhibits the predicted geometric degeneration, that VaDE geometry diagnostics predict future component loss, and that real-data injection stabilizes both distributional and geometric collapse metrics.

1 Introduction

Recursive training on synthetic data can cause generative models to lose rare modes, shrink tail mass, and drift from the original distribution [Shumailov et al., 2024, Bertrand et al., 2024, Dohmatob et al., 2024]. Existing explanations are mainly distributional, through variance shrinkage, tail loss, and divergence across generations [Shumailov et al., 2024, Alemohammad et al., 2024, Hataya et al., 2023], or probabilistic, through multinomial and Wright–Fisher absorption dynamics [Suresh et al., 2024, Borkar, 2025, Jangjoo et al., 2025, Benati et al., 2025, Riis, 2026]. We ask a complementary question: what happens to the *loss geometry* when recursive training eliminates modes?

Our thesis is that collapse is also a singularization process. When a mode disappears, its associated parameters become locally unidentifiable; before exact disappearance, small component weight already induces $O(\pi_k)$-scale Fisher curvature in component-specific directions. In a labeled Gaussian mixture abstraction, we show that mode extinction induces a monotone trajectory in the real log-canonical threshold (RLCT), the singular-learning-theory measure of effective complexity [Watanabe, 2009, 2018]. In VaDE-style neural latent-variable models [Jiang et al., 2017], we show that the same

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mechanism appears as continuous latent-prior Fisher-curvature collapse, while shared decoder curvature is controlled by latent feature covariance. These results motivate loss-geometry diagnostics based on latent-prior Fisher curvature, local learning coefficient estimates [Furman and Lau, 2024], Hessian and Gauss–Newton spectra, decoder-block curvature, feature-rank concentration, and distributional mode coverage.

Contributions. We contribute: **(i)** an RLCT trajectory theorem for labeled Gaussian-mixture recursion, plus a finite-horizon real-data protection condition $\alpha \geq \log(KG/\delta)/(n\pi^*_{\min})$; **(ii)** a neural latent-variable bridge showing that near-extinct VaDE components have $O(\pi_k)$-scale Fisher curvature, with exact Fisher nullity at extinction, and a decoder curvature identity linking final-layer Hessian rank to latent feature covariance; and **(iii)** empirical validation across Gaussian mixtures, VaDE, standard VAEs, and a compact DDPM, where synthetic-only recursion induces geometry degeneration, VaDE geometry diagnostics predict future component loss, and real-data injection stabilizes distributional and geometric collapse metrics.

Related work. Prior work on model collapse characterizes recursive training through distributional degradation, such as tail loss, variance shrinkage, and divergence across generations, or through probabilistic absorption dynamics in multinomial and Wright–Fisher-style processes. Separately, singular learning theory uses the RLCT and local learning coefficient to quantify effective complexity in singular models [Watanabe, 2009, 2018, Furman and Lau, 2024]. Our work connects these literatures by tracking an RLCT trajectory along recursive collapse and by showing that near-extinct latent components induce quantitatively small Fisher curvature before exact singularity is reached. This turns singular geometry into a pre-collapse diagnostic rather than only a post-extinction description. Full related work appears in the appendix.

2 Theory

2.1 Labeled Gaussian mixture recursion

We analyze a K-component unit-variance Gaussian mixture

$$p_\theta(x) = \sum_{k=1}^{K} \pi_k \varphi(x \mid \eta_k), \qquad \theta = (\pi, \eta_1, \ldots, \eta_K),$$

where $\varphi(\cdot \mid \eta)$ is the unit-variance Gaussian density with mean η. The initial parameter θ^* satisfies $\pi^*_k > 0$ for all k and has distinct means. Starting from $\theta_0 = \theta^*$, the labeled recursion draws

$$z_i^{(g)} \sim \mathrm{Categorical}(\pi_g), \qquad x_i^{(g)} \mid z_i^{(g)} = k \sim \varphi(\cdot \mid \eta_{g,k}).$$

Let $N_{g,k} = |\{i : z_i^{(g)} = k\}|$. The labeled MLE update is

$$\pi_{g+1,k} = \frac{N_{g,k}}{n}, \qquad \eta_{g+1,k} = \frac{1}{N_{g,k}} \sum_{i : z_i^{(g)} = k} x_i^{(g)} \quad (N_{g,k} \geq 1),$$

with $\eta_{g+1,k}$ irrelevant when $N_{g,k} = 0$. Let

$$E_g = \{k : \pi_{g,k} = 0\}, \qquad r_g = K - |E_g|.$$

Then $\pi_{g+1} = N_g/n$, $N_g \sim \mathrm{Multinomial}(n, \pi_g)$, is the neutral K-allele Wright–Fisher chain. We use its standard martingale, monotone-extinction, and vertex-absorption properties. The trajectory RLCT below is evaluated in the separated-survivor regime of the Aoyagi–Nagata formula; finite-sample peak coalescence is not the extinction-driven singularization mechanism studied here.

2.2 The singular complexity trajectory

For a K-component univariate Gaussian mixture fit to a target with r separated true peaks, the Aoyagi–Nagata RLCT is

$$\lambda(K, r) = (r - 1) + \frac{i^2 + i + 2(K - r + 1)}{4(i + 1)}, \qquad i = \max\{j \in \mathbb{Z}_{\geq 0} : j^2 + j \leq 2(K - r + 1)\}.$$

It satisfies

$$\lambda(K,K) = \frac{2K-1}{2}, \qquad \lambda(K,r) - \lambda(K,r-1) \geq \frac{3}{4} \quad (2 \leq r \leq K),$$

with the closed-form drop proved in Appendix D. Define the trajectory RLCT by

$$\lambda_g := \lambda(K, r_g).$$

We write

$$I(\theta) := \mathbb{E}_{x \sim p_\theta}\left[\nabla_\theta \log p_\theta(x) \nabla_\theta \log p_\theta(x)^\top\right]$$

for the Fisher information.

Theorem 1 (Singular complexity trajectory in labeled Gaussian mixture recursion)**.** *In the labeled synthetic-only recursion above, λ_g is non-increasing almost surely. Whenever a component becomes extinct, λ_g drops by at least $3/4$. The process is eventually absorbed at a single surviving component, so $\lambda_g \to \lambda(K,1)$. Moreover,*

$$\dim\ker I(\theta_g) \geq |E_g|, \qquad \liminf_{g\to\infty} \dim\ker I(\theta_g) \geq K-1.$$

The proof combines monotone extinction in the Wright–Fisher chain, monotonicity of the Aoyagi–Nagata RLCT, and the fact that parameters of extinct components have zero score; see Appendix.

Proposition 1 (Finite-horizon prevention under real-data injection)**.** *In the labeled α-mixed recursion, each generation uses $\lfloor(1-\alpha)n\rfloor$ labeled synthetic samples from p_{θ_g} and $\lceil \alpha n \rceil$ labeled real samples from $p^* = p_{\theta^*}$. Let $N_{g,k}^{(\alpha)}$ be the total number of mixed training samples assigned to component k, and assume $\pi_k^* \geq \pi_{\min}^* > 0$. Then for any horizon G,*

$$\mathbb{P}\Big(\exists g < G, \exists k : N_{g,k}^{(\alpha)} = 0\Big) \leq KG \exp\{-\alpha n \pi_{\min}^*\}.$$

Consequently, if

$$\alpha \geq \frac{\log(KG/\delta)}{n\pi_{\min}^*},$$

then with probability at least $1-\delta$, no component becomes extinct over the horizon and

$$\lambda_g = \lambda(K,K) = \frac{2K-1}{2}, \qquad 0 \leq g \leq G.$$

The proof, including the sharper $m_\alpha = \lceil \alpha n \rceil$ bound, is given in Appendix E.2.

2.3 From mixtures to neural generative models

The mixture theorem identifies the exact singularity at extinction. For VaDE, with latent dimension d, Gaussian-mixture latent prior $\pi_c \mathcal{N}(z \mid \mu_c, \Sigma_c)$, and decoder $p_\psi(x \mid z)$, define the component marginal

$$q_c(x) := \int p_\psi(x \mid z) \mathcal{N}(z \mid \mu_c, \Sigma_c)\, dz, \qquad p_\Theta(x) = \sum_{c=1}^K \pi_c q_c(x), \qquad \rho_c(x) = \frac{\pi_c q_c(x)}{p_\Theta(x)}.$$

For a latent-prior parameter block $a_c \in \{\mu_c, \Sigma_c\}$,

$$\nabla_{a_c} \log p_\Theta(x) = \rho_c(x) \nabla_{a_c} \log q_c(x).$$

Thus, if

$$\mathbb{E}_{x\sim q_c}\left[\|\nabla_{a_c} \log q_c(x)\|^2\right] \leq C_c,$$

then the corresponding Fisher block satisfies

$$\operatorname{tr} I_{a_c a_c}(\Theta) \leq \pi_c C_c.$$

Near-extinct components therefore induce $O(\pi_c)$-scale Fisher curvature before exact extinction. If $E = \{c : \pi_c = 0\}$, then the corresponding latent-prior scores vanish exactly and

$$\dim\ker I(\Theta) \geq |E| \left(d + \frac{d(d+1)}{2}\right).$$

Shared decoder parameters need not become Fisher-null. However, for a final decoder layer $f_W(z) = W\phi(z)$ with reconstruction loss

$$\mathcal{L}_g(W) = \frac{1}{2\sigma^2}\mathbb{E}_{(z,x)\sim Q_g}\|W\phi(z) - x\|_2^2,$$

where Q_g is the generation-g latent-data distribution and I_{out} is the output-dimensional identity matrix,

$$\nabla^2_{\text{vec}(W)}\mathcal{L}_g(W) = \frac{1}{\sigma^2}\mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top] \otimes I_{\text{out}}.$$

Thus decoder curvature can degenerate when recursive training concentrates latent features.

These results motivate four diagnostics: VaDE latent-prior Fisher collapse (P1), empirical LLC decay (P2), Hessian/Gauss–Newton and feature-rank concentration (P3), and stabilization under real-data injection (P4). For standard VAEs, (P2)–(P4) are empirical signatures; for DDPMs, the applicable signatures are the broader rank and stabilization diagnostics (P3)–(P4).

3 Experiments

The experiments test predictions (P1)–(P4) across four tiers of increasing model complexity: univariate Gaussian mixtures, where Theorem 1 applies exactly; VaDE, where the latent-prior Fisher-scaling and decoder-curvature predictions can be tested directly; standard VAEs, where the labeled-mixture abstraction is absent but latent-variable geometry remains; and a compact DDPM, which tests whether broader geometry-collapse signatures persist beyond latent-variable models. Across tiers, the α-mixed recursion serves as a counterfactual control: synthetic-only training ($\alpha = 0$) is the regime in which Theorem 1 predicts singular-complexity descent, while Proposition 1 gives a sufficient finite-horizon condition under which real-data injection prevents RLCT descent in the labeled mixture tier. The neural experiments test whether the same stabilizing effect persists in VaDE, standard VAEs, and DDPMs.

3.1 Setup and diagnostics

Across tiers, we run recursive training

$$\mathcal{D}_{g+1} \sim p_{\theta_g}^{\otimes n}, \qquad \theta_{g+1} = \text{Train}(\mathcal{D}_{g+1}, \theta_g),$$

with tier-specific training rules. For Gaussian mixtures, Train is labeled MLE; for VaDE and standard VAEs, it is ELBO maximization; and for the DDPM tier, it is denoising-score training on the generated image set. We compare synthetic-only recursion to an α-mixed control that replaces an α-fraction of each generation's samples with fresh samples from the original distribution.

We track both distributional and geometric collapse metrics. Distributional diagnostics include active component count for mixtures and VaDE, class-coverage entropy for image models, held-out negative log-likelihood where available, and FID for the DDPM tier. Geometry diagnostics include empirical LLC estimates, Hessian and Gauss–Newton spectral concentration, componentwise latent-prior Fisher curvature and near-nullity for VaDE, decoder feature-covariance effective rank, and DDPM U-Net feature-covariance and blockwise Gauss–Newton effective ranks. We also evaluate whether these diagnostics predict future collapse events, rather than only tracking collapse after it occurs. Unless otherwise stated, curves report mean $\pm$ one standard deviation over 10 seeds; the DDPM tier uses 5 seeds. Definitions, estimator hyperparameters, architectures, compute details, and bootstrap procedures are given in Appendix.

3.2 Tier 1 — Gaussian mixtures: exact validation of the theory

Question: *Does the labeled-mixture theorem hold exactly in its intended setting?*

Tier 1 validates the exact labeled-mixture mechanism: synthetic-only recursion produces absorbing support loss, stepwise RLCT descent, and increasing Fisher nullity, while real-data injection prevents extinction over the finite horizon. Full trajectories, relaxation-time scaling, and LLC calibration are shown in Appendix.

3.3 Tier 2 — VaDE: latent-prior singularization in a neural model

Question: *Does the latent-prior curvature-collapse mechanism survive in a neural latent-variable model?*

We train VaDE on MNIST with $K = 10$ latent components, latent dimension $d = 10$, $n = 5000$ generated samples per generation, $G = 50$ generations, 10 seeds, and $\alpha \in \{0, 0.1, 0.25, 0.5\}$. Architecture and optimization details are given in Appendix. Components with $\pi_{g,k} < 10^{-3}$ are treated as near-extinct:

$$E_g^\varepsilon = \{k : \pi_{g,k} < 10^{-3}\}, \qquad r_g^\varepsilon = K - |E_g^\varepsilon|.$$

Figure 1 shows that the latent-prior singularization mechanism persists in VaDE, despite training without oracle component labels. Under synthetic-only recursion, the active component count falls from 10.0 to 3.6 ± 0.8 by generation 50, while real-data injection preserves latent support: $\alpha = 0.25$ and $\alpha = 0.5$ retain 8.9 ± 0.6 and 9.7 ± 0.3 active components, respectively.

The Fisher geometry matches the VaDE bridge. For each component k, define

$$\kappa_{g,k} := \frac{1}{d + d(d+1)/2} \operatorname{tr} I_{(\mu_{g,k}, \Sigma_{g,k}),(\mu_{g,k}, \Sigma_{g,k})}(\Theta_g).$$

Section 2.3 predicts $\kappa_{g,k} = O(\pi_{g,k})$; across generations, components, and seeds, a log-log regression gives slope 0.94 with $R^2 = 0.85$. Thresholding at $\pi_{g,k} < 10^{-3}$, each near-extinct full-covariance component contributes

$$d + \frac{d(d+1)}{2} = 65$$

near-null directions; at generation 50, synthetic-only recursion produces 416 ± 52 near-null Fisher directions, matching the predicted count $|E_g^\varepsilon| \cdot 65$.

The remaining neural diagnostics degenerate in parallel. Under $\alpha = 0$, decoder feature-covariance effective rank decreases from 8.9 ± 0.4 to 3.9 ± 0.7, empirical LLC decreases from 128.4 ± 6.7 to 80.1 ± 8.9, and Hessian/Gauss–Newton effective ranks fall to roughly half their initial values. Under $\alpha \geq 0.25$, latent support, feature rank, and aggregate geometry diagnostics remain substantially closer to their initial values. Thus VaDE supports (P1)–(P4): latent-prior Fisher-curvature collapse, near-nullity, complexity decay, spectral concentration, and stabilization under real-data injection.

3.4 Tier 3 — Standard VAEs: empirical signatures without explicit mixtures

Question: *Do analogous singular-geometric signatures appear without an explicit mixture structure?*

We train a standard VAE on MNIST with a single Gaussian latent prior, latent dimension $d = 10$, and the same MLP encoder/decoder architecture as Tier 2. Each recursive generation uses $n = 5000$ generated samples and 50 epochs of continuation training. We run $G = 50$ generations, 10 seeds, and $\alpha \in \{0, 0.1, 0.25, 0.5\}$. Since the model has no explicit mixture weights, we evaluate collapse using distributional mode coverage and loss-geometry diagnostics rather than latent-component extinction.

Figure 2 tests whether the signatures predicted by the mixture and VaDE analyses persist in a standard VAE. Under synthetic-only recursion, class-coverage entropy decreases from 2.30 ± 0.03 to 1.21 ± 0.18, indicating loss of generated-sample diversity across MNIST classes. The empirical LLC, using the estimator calibrated in Tier 1, decreases from 116.2 ± 5.8 to 69.4 ± 7.6, supporting (P2). Hessian and Gauss–Newton effective ranks decrease to roughly one half of their initial values, showing spectral concentration of the loss landscape and supporting (P3). The decoder feature-covariance effective rank decreases from 8.6 ± 0.5 to 3.7 ± 0.6, indicating that the decoder-side degeneration observed in VaDE persists without an explicit mixture prior. Real-data injection stabilizes all diagnostics: for $\alpha \geq 0.25$, class-coverage entropy remains above 2.05, LLC remains above 98.0, and decoder feature rank remains above 7.4 at generation 50. Thus the standard VAE experiment supports (P2)–(P4) in a model where the exact latent-prior nullity mechanism is absent.

3.5 Tier 4 — DDPM: empirical signatures beyond latent-variable models

Question: *Do geometry-collapse signatures persist in a generative model without an explicit mixture prior or VAE-style latent decoder?*

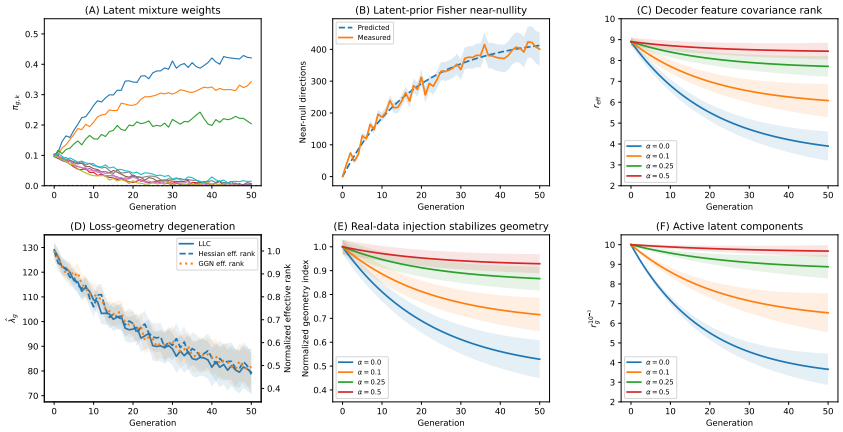


Figure 1: **VaDE exhibits latent-prior singularization under recursive training.** (A) Latent mixture weights show near-extinction under synthetic-only recursion. (B) Latent-prior Fisher near-nullity tracks the predicted count $|E_g^\varepsilon|(d + d(d+1)/2)$, with $d = 10$. (C) Decoder feature-covariance effective rank decreases as latent support collapses. (D) Empirical LLC, Hessian effective rank, and Gauss–Newton effective rank decrease under synthetic-only recursion. (E) Real-data injection stabilizes aggregate geometry diagnostics. (F) Active latent component count shows that $\alpha \geq 0.25$ largely preserves latent support. MNIST, $K = 10$, $d = 10$, $n = 5000$, $G = 50$, mean over 10 seeds with 1σ bands.

To test whether the observed signatures are specific to latent-variable models, we run recursive training with a compact denoising diffusion probabilistic model (DDPM) on FashionMNIST. Unlike VaDE, the DDPM has no explicit latent mixture weights, and unlike the VAE, it does not use a latent decoder block matching the final-layer curvature identity. Thus this tier does not test the exact latent-prior nullity prediction (P1). Instead, it tests whether the broader empirical signatures, distributional degradation, feature-rank concentration, loss-geometry rank concentration, and real-data stabilization, persist outside the mixture/VAE setting.

We train the DDPM under the same recursive protocol with $\alpha \in \{0, 0.1, 0.25\}$ over $G = 20$ recursive generations and report mean over 5 seeds. Distributional collapse is measured by class-coverage entropy and FID. Geometry is measured by the effective rank of U-Net feature covariances and a blockwise empirical Gauss–Newton effective rank computed on the denoising loss. Figure 3 shows that synthetic-only recursion produces both distributional and geometric collapse. At generation $G = 20$, class-coverage entropy decreases to 1.37 ± 0.05, FID increases to 90.0 ± 2.3, U-Net feature effective rank decreases to 0.51 ± 0.02, and blockwise Gauss–Newton effective rank decreases to 0.53 ± 0.04. Real-data injection substantially stabilizes all four trajectories: with $\alpha = 0.25$, the corresponding final values are 2.28 ± 0.05, 29.9 ± 1.0, 0.96 ± 0.02, and 0.89 ± 0.03. Thus, in this FashionMNIST DDPM experiment, the broader geometry-collapse signatures persist beyond VAE-style latent-variable models, although the exact mixture-prior nullity mechanism remains specific to VaDE.

3.6 Real-data injection and robustness

Across the GMM, VaDE, and VAE tiers, real-data injection acts as a consistent stabilizing control. Figure 5A summarizes the final normalized collapse index across Gaussian mixtures, VaDE, and standard VAEs as a function of α. Synthetic-only recursion produces the strongest collapse in every tier, with final collapse indices 0.82 ± 0.07, 0.74 ± 0.08, and 0.69 ± 0.07 for GMM, VaDE, and VAE, respectively. Increasing α suppresses collapse monotonically, and $\alpha \geq 0.25$ reduces the mean final collapse index to approximately 0.2 or below in all three settings. Together with the DDPM results in Section 3.5, this shows that the stabilizing effect of real data is not specific to the labeled mixture abstraction or to VAE-style latent-variable models.

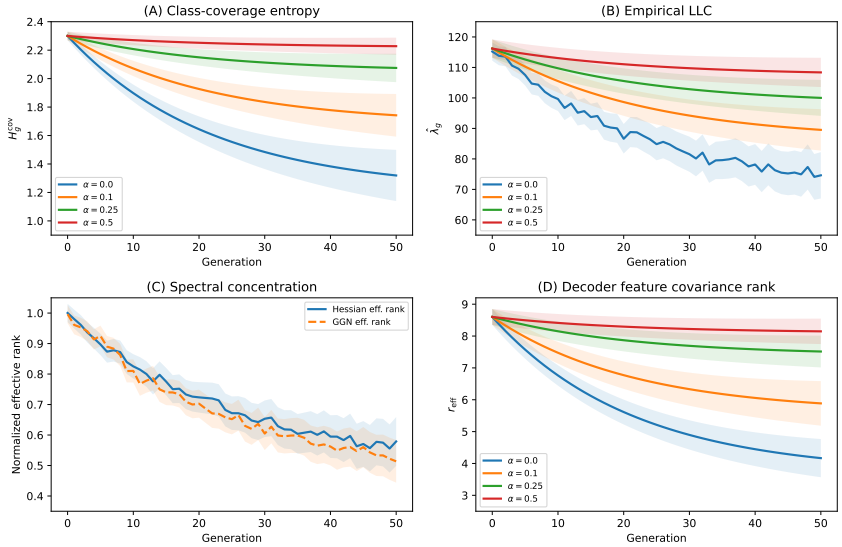


Figure 2: **Standard VAEs exhibit singular-geometric collapse signatures without explicit mixture structure.** (A) Class-coverage entropy decreases under synthetic-only recursion, showing distributional mode loss. (B) Empirical LLC decreases across generations, supporting complexity decay. (C) Hessian and Gauss–Newton effective ranks decrease, showing spectral concentration. (D) Decoder feature-covariance effective rank decreases under synthetic-only recursion and stabilizes under real-data injection. MNIST, latent dimension $d = 10$, $n = 5000$, $G = 50$, mean over 10 seeds with 1σ bands.

We next test whether the observed singular-geometric signatures depend on diagnostic choices and whether the quantitative near-extinction prediction is visible at the component level. Figure 5B tests the central scaling law from Section 2.3: componentwise VaDE latent-prior Fisher curvature decreases with mixture weight, with log-log slope 0.94 and $R^2 = 0.85$. This supports the predicted $O(\pi_{g,k})$ Fisher-curvature collapse before exact component extinction. Figure 5C shows that VaDE latent-prior near-nullity remains aligned with the predicted trajectory across near-extinction thresholds $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}\}$. Figure 5D shows that the LLC decay under synthetic-only recursion is stable across SGLD estimator settings, including doubled chain length and step-size perturbations. Figure 5E shows that decoder feature-covariance rank collapse persists across latent dimensions $d \in \{5, 10, 20\}$. Thus the main conclusions are robust to the threshold defining near-extinction, the LLC estimator configuration, and moderate architecture changes, while the componentwise Fisher-scaling test supports the near-nullity mechanism before exact extinction.

Early-warning prediction. Finally, we test whether the proposed geometric diagnostics predict future collapse rather than only describe collapse after it occurs. In VaDE, we define a future-collapse event at generation g by

$$\mathcal{C}_{g,\Delta} := \mathbf{1}\{r^{\varepsilon}_{g+\Delta} < r^{\varepsilon}_{g}\}, \qquad \Delta = 10,$$

so that $\mathcal{C}_{g,\Delta} = 1$ indicates that at least one latent component becomes near-extinct within the next ten recursive generations. We train simple logistic predictors using only distributional metrics, only geometric metrics, or both. Distribution-only predictors include current active component count, class-coverage entropy, and held-out negative log-likelihood. Geometry-only predictors include componentwise latent-prior Fisher curvature, Fisher near-nullity, empirical LLC, Hessian and Gauss–Newton effective rank, and decoder feature-covariance rank. We evaluate prediction using leave-one-seed-out splits, so checkpoints from the same recursive trajectory never appear in both train and test sets. This prevents the classifier from exploiting trajectory-specific temporal correlations rather than learning transferable early-warning signals.

Figure 4 shows that geometry-only predictors achieve AUROC 0.85 ± 0.03, outperforming the distribution-only baseline at 0.72 ± 0.04, while the combined predictor reaches 0.95 ± 0.01. Thus the

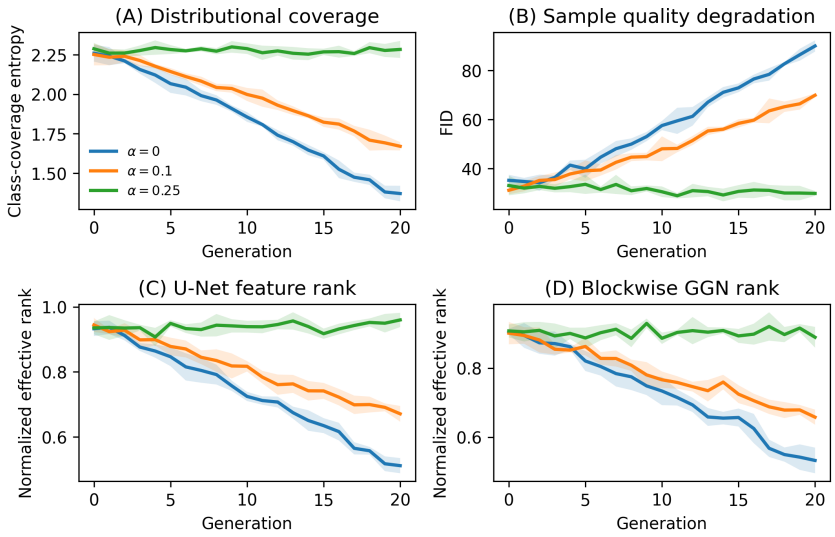


Figure 3: **DDPM recursion exhibits geometry-collapse signatures beyond latent-variable models.** A compact DDPM is recursively trained on FashionMNIST with $\alpha \in \{0, 0.1, 0.25\}$. Under synthetic-only recursion ($\alpha = 0$), class-coverage entropy decreases, FID increases, U-Net feature-covariance effective rank decreases, and blockwise Gauss–Newton effective rank decreases. Real-data injection stabilizes both distributional and geometric diagnostics, with $\alpha = 0.25$ largely preserving coverage, sample quality, feature rank, and curvature rank over $G = 20$ generations. Mean over 5 seeds with 1σ bands.

proposed loss-geometry diagnostics provide early-warning information about future collapse beyond contemporaneous distributional coverage.

Table 1: **Summary of predictions across the experimental ladder.** GMM validates the exact mixture theorem, VaDE tests the neural latent-prior bridge, standard VAEs test empirical signatures without explicit mixture structure, and DDPM tests whether broader geometry-collapse signatures persist beyond latent-variable models.

Prediction	Status	GMM	VaDE	VAE	DDPM
(P1) Fisher curvature collapse / nullity	$O(\pi_k)$ scaling; exact at extinction	✓	✓	n/a	n/a
(P2) Singular complexity decreases	Exact RLCT / empirical LLC	✓	✓	✓	n/a
(P3) Spectral and feature-rank concentration	Empirical, curvature-guided	✓	✓	✓	✓
(P4) Real-data stabilization	Finite-horizon theory / empirical control	✓	✓	✓	✓
(P5) Early-warning prediction	Held-out-seed prediction of future component loss	n/a	✓	n/a	n/a

4 Discussion and limitations

This work reframes recursive model collapse as a loss-geometry phenomenon. In labeled mixtures, mode extinction induces RLCT descent and Fisher nullity; in VaDE, near-extinct components already produce $O(\pi_k)$-scale Fisher curvature before exact extinction. Standard VAEs and a compact DDPM do not have the exact mixture-prior mechanism, but still show LLC decay, spectral concentration, feature-rank collapse, and stabilization under real-data injection.

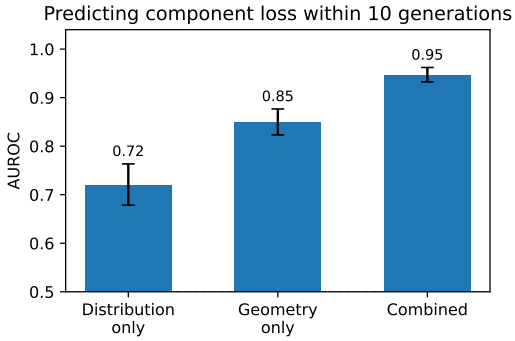


Figure 4: **Loss-geometry diagnostics predict future collapse.** Using VaDE checkpoints, we predict whether the active latent-component count r_g^{ε} will decrease within the next $\Delta = 10$ recursive generations. Prediction is evaluated with leave-one-seed-out splits, so checkpoints from the same recursive trajectory do not appear in both train and test sets. Geometry-only predictors outperform distribution-only predictors, and combining both gives the strongest early-warning performance. Bars show mean AUROC over held-out-seed folds with 1σ error bars.

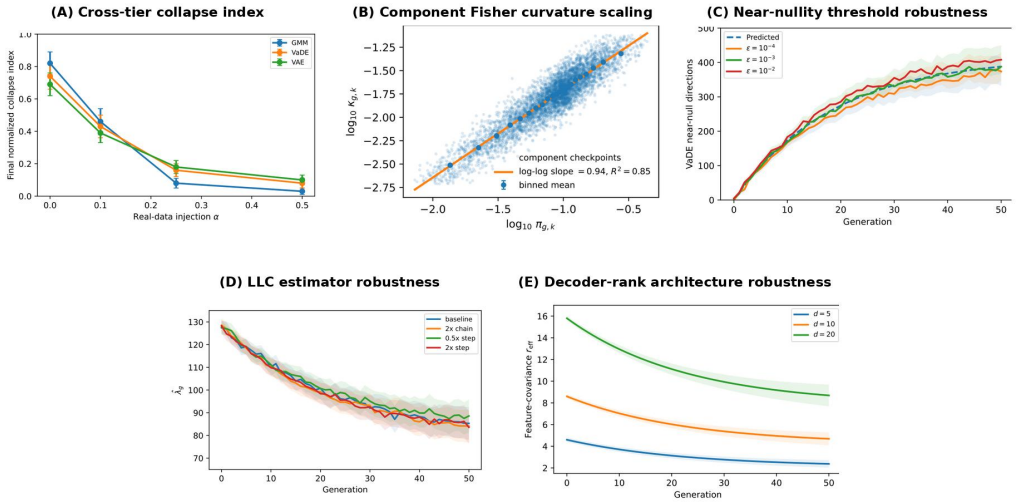


Figure 5: **Robustness of singular-geometric collapse diagnostics.** (A) Final normalized collapse index across GMM, VaDE, and standard VAE decreases as real-data injection α increases. (B) Componentwise VaDE latent-prior Fisher curvature scales with mixture weight, with log-log slope 0.94 and $R^2 = 0.85$, supporting the $O(\pi_{g,k})$ near-extinction prediction of Section 2.3. (C) VaDE latent-prior near-nullity is stable across near-extinction thresholds ε. (D) LLC decay under synthetic-only recursion is stable across SGLD estimator settings. (E) Decoder feature-covariance rank collapse persists across latent dimensions. All panels report mean over 10 seeds with 1σ bands where applicable.

Real-data injection gives both an empirical control and, in the labeled mixture abstraction, a finite-horizon guarantee: Proposition 1 shows that $\alpha = \Omega(\log(KG/\delta)/(n\pi^*_{\min}))$ prevents RLCT descent over horizon G with probability at least $1 - \delta$. The main limitations are that the neural experiments are small-scale, early warning is demonstrated only for VaDE component loss, and LLC/Hessian/Gauss–Newton diagnostics are more expensive than standard sample-quality metrics. Scaling these diagnostics to larger diffusion, autoregressive, and multimodal models remains future work.

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A Related work

Model collapse in recursive generative training. Shumailov et al. [2024] demonstrated that recursive training of generative models on their own outputs causes progressive distributional degradation, with rare modes lost first and tail mass disappearing across generations. Subsequent work has established this phenomenon across model classes: Alemohammad et al. [2024] for self-consuming generative loops, Hataya et al. [2023] for image-generation pipelines, Bertrand et al. [2024] for iterative retraining stability, and Dohmatob et al. [2024] for the regression case. These accounts characterize collapse *distributionally*, through variance shrinkage, tail loss, and divergence metrics. A second line of work analyzes the underlying *probabilistic dynamics*: Suresh et al. [2024] establish first-extinction rates for $K = 2$ multinomial recursion, Borkar [2025] prove vertex absorption for general categorical chains, Jangjoo et al. [2025] treat the Random Energy Model case, Benati et al. [2025] derive Wright–Fisher first-extinction laws, and Riis [2026] observe that the same dynamics describe recursive n-gram language model retraining. Our work builds on this dynamical foundation but addresses a different question: not what extinction does to the data distribution or the mixture-weight chain, but what it does to the singular geometry of the parameter space. The closed-form RLCT trajectory of Theorem **??** characterizes a complexity quantity that no prior collapse paper computes.

Singular learning theory and the local learning coefficient. Singular learning theory [Watanabe, 2009, 2018] provides the framework in which the real log-canonical threshold quantifies effective parameter complexity for singular models, including neural networks. Aoyagi [2010] and Aoyagi and Nagata [2012] compute closed-form RLCT formulas for normal mixture models via Vandermonde matrix-type singularities, with extensions to other regular exponential families by Yamazaki et al. [2010]. The local learning coefficient (LLC) of Lau et al. [2025], with the scalable estimator of Furman and Lau [2024], makes the RLCT empirically measurable for deep networks via stochastic-gradient Langevin dynamics. Recent applications include Hoogland et al. [2024] on developmental phase transitions in transformer training, Chen et al. [2023] on toy models of superposition, and Wang et al. [2024] on attention-head specialization. To our knowledge, no prior work has applied SLT to recursive generative training, nor connected the LLC to model-collapse dynamics. Our Theorem **??** provides the first closed-form RLCT trajectory along a recursive-training process; Propositions **??**–**??** extend the unidentifiability mechanism to neural latent-variable models.

Population-genetics absorption times. The Wright–Fisher diffusion provides the classical setting in which absorption times of multinomial drift processes are computed. The bi-allelic case is due to Kimura [1955], with the multi-allelic generalization treated systematically in Ewens [2004, Chapter 5]. **?** establish the entropy-dissipation rate for K-allele neutral Wright–Fisher diffusion, which we apply to obtain the complexity-relaxation rate of Proposition **??**. The connection between recursive generative training and Wright–Fisher dynamics is implicit in Benati et al. [2025] and Riis [2026], but the rate result for complexity relaxation has not been previously stated; this connection is what links per-generation sample size n to the expected number of generations until full singular collapse.

B Proof of the RLCT trajectory theorem

Proof of Theorem 1. In the labeled recursion, if component k is extinct at generation g, then $\pi_{g,k} = 0$. Hence $N_{g,k} = 0$ almost surely at the next generation and $\pi_{g+1,k} = 0$. Therefore $E_g \subseteq E_{g+1}$ and $r_{g+1} \leq r_g$.

By the Aoyagi–Nagata monotonicity property proved in Appendix D, $\lambda(K, r)$ is strictly increasing in the number r of separated active peaks, with

$$\lambda(K, r) - \lambda(K, r-1) \geq \frac{3}{4}.$$

Since $\lambda_g = \lambda(K, r_g)$ and r_g is non-increasing, λ_g is non-increasing almost surely. If a new component becomes extinct, then $r_{g+1} < r_g$, so the RLCT drops by at least $3/4$.

The weight process is the neutral finite-population Wright–Fisher chain

$$\pi_{g+1} = N_g/n, \qquad N_g \sim \text{Multinomial}(n, \pi_g).$$

Its absorbing states are the simplex vertices. Starting from strictly positive weights, the chain is absorbed at a vertex almost surely, so eventually exactly one component survives. Hence $r_g \to 1$ and

$$\lambda_g \to \lambda(K, 1).$$

Finally, if $k \in E_g$, then the model density at generation g no longer depends on the component mean η_k. Therefore

$$\partial_{\eta_k} \log p_{\theta_g}(x) = 0$$

for all x in the support of the surviving mixture, so the η_k-coordinate is a Fisher-null direction. Thus

$$\dim \ker I(\theta_g) \geq |E_g|.$$

Since the Wright–Fisher chain is eventually absorbed at a single surviving component, $|E_g| \to K-1$, and therefore

$$\liminf_{g\to\infty} \dim \ker I(\theta_g) \geq K - 1.$$

□

C Setup and notation

Univariate Gaussian mixture model. For the main theorem we analyze a K-component univariate Gaussian mixture with known unit variance:

$$p_\theta(x) \;=\; \sum_{k=1}^{K} \pi_k \, \phi(x \mid \eta_k), \qquad \phi(x \mid \eta) := \tfrac{1}{\sqrt{2\pi}} \exp\!\big(-\tfrac{1}{2}(x-\eta)^2\big), \tag{1}$$

parameterized by mixture weights $\pi \in \Delta^{K-1} := \{\pi \in \mathbb{R}^K_{\geq 0} : \sum_k \pi_k = 1\}$ and component means $\eta_k \in \mathbb{R}$. The full parameter is $\theta = (\pi, \eta_1, \ldots, \eta_K) \in \Theta := \Delta^{K-1} \times \mathbb{R}^K$, with $D := \dim \Theta = 2K - 1$. The data-generating parameter is $\theta^* = (\pi^*, \eta_1^*, \ldots, \eta_K^*)$ with $p^* := p_{\theta^*}$. The model is identifiable up to component relabeling [Titterington et al., 1985]; we work on the quotient $\Theta/\mathfrak{S}_K$, and all quantities below are invariant under this action.

Remark 1 (General exponential-family extension). The univariate Gaussian-known-variance setting is the regime in which Aoyagi and Nagata [2012] provide a closed-form RLCT formula for normal mixtures. Generalization to higher data dimension and to other regular exponential families is discussed in Remark **??**.

Assumption 1 (Non-degenerate data generator)**.** *$\pi_k^* > 0$ for all k and $\eta_j^* \neq \eta_k^*$ for all $j \neq k$.*

Recursive synthetic-only training (labeled abstraction). Our theoretical analysis treats a *labeled* (oracle-separated) recursion, in which generated samples retain their latent component identity. This abstraction isolates the singular-geometric mechanism from the unlabeled-mixture identifiability problem, and matches the abstraction used implicitly by Suresh et al. [2024], Borkar [2025], Jangjoo et al. [2025], Benati et al. [2025] in their analyses of recursive multinomial dynamics. The neural experiments of Section 3 test whether the same singular-geometric signatures persist in unlabeled, end-to-end-trained generative models.

Fix sample size $n \in \mathbb{N}$ per generation. Starting from $\theta_0 = \theta^*$, define $\{\theta_g\}_{g\geq 0}$ as follows. At generation g, draw labeled synthetic samples

$$(x_i^{(g)}, z_i^{(g)})_{i=1}^n \overset{\text{i.i.d.}}{\sim} p_{\theta_g}, \qquad z_i^{(g)} \in \{1, \ldots, K\}, \tag{2}$$

where $z_i^{(g)} \mid \theta_g \sim \mathrm{Categorical}(\pi_g)$ is the latent component identity and $x_i^{(g)} \mid z_i^{(g)} = k, \theta_g \sim \phi(\cdot \mid \eta_{g,k})$. Let $N_{g,k} := |\{i : z_i^{(g)} = k\}|$ denote the count of samples drawn from component k, so that $(N_{g,1}, \ldots, N_{g,K}) \mid \theta_g \sim \mathrm{Multinomial}(n, \pi_g)$. The next-generation parameter is the labeled-MLE update

$$\pi_{g+1,k} \;=\; \frac{N_{g,k}}{n}, \qquad \eta_{g+1,k} \;=\; \frac{1}{N_{g,k}} \sum_{i : z_i^{(g)} = k} x_i^{(g)} \quad \text{whenever } N_{g,k} \geq 1, \tag{3}$$

with $\eta_{g+1,k}$ left undefined when $N_{g,k} = 0$ (the value is irrelevant since $\pi_{g+1,k} = 0$). Let $\mathcal{F}_g = \sigma(\{(x_i^{(j)}, z_i^{(j)})\}_{i,j<g})$ be the natural filtration.

Extinct set. For generation g, define the *extinct set* $E_g := \{k \in \{1, \dots, K\} : \pi_{g,k} = 0\}$ and the *support* $S_g := \{1, \dots, K\} \setminus E_g$. We say component k is *extinct at generation* g if $k \in E_g$ and *surviving* otherwise.

Fisher information and loss-landscape diagnostics. The Fisher information at $\theta \in \Theta$ is

$$I(\theta) \;=\; \mathbb{E}_{x \sim p_\theta}\left[\nabla_\theta \log p_\theta(x)\, \nabla_\theta \log p_\theta(x)^\top\right] \in \mathbb{R}^{D \times D}. \tag{4}$$

Mixture models are singular on the boundary stratum $\partial_{\deg}\Theta := \{\theta : \min_k \pi_k = 0\}$ [Watanabe, 2009, Aoyagi, 2010]. For sample $X = (x_1, \dots, x_m)$ we write $\mathcal{L}(\theta; X) := -m^{-1} \sum_i \log p_\theta(x_i)$. Our three diagnostics at θ_g are: the empirical Hessian $H_g := \nabla^2 \mathcal{L}(\theta_g; X^{(g)})$ and its spectrum; the Gauss–Newton matrix G_g (defined when used); and the real log-canonical threshold λ_g [Watanabe, 2009] with empirical estimator $\hat{\lambda}_g$ (the local learning coefficient of Lau et al. [2025], computed via SGLD as in Furman and Lau [2024]). Hessian spectra and the RLCT are not equivalent for singular models; we treat them as complementary.

Real-data injection. The α-mixed recursion replaces $X^{(g)}$ in (2) with $\lfloor (1-\alpha)n \rfloor$ labeled samples from p_{θ_g} and $\lceil \alpha n \rceil$ labeled samples from p^*, drawn independently; $\alpha = 0$ is synthetic-only and $\alpha = 1$ is standard MLE from p^*.

C.1 Background: recursive multinomial dynamics

The mixture-weight component of the recursion (3) is a finite-state Markov chain on the rational simplex $\Delta_n^{K-1} := \{\pi \in \Delta^{K-1} : n\pi_k \in \mathbb{N}_0\}$. The following facts are established in the recent recursive-training literature; we record them here as background and use them as scaffolding for the main results in the next subsection.

Lemma 1 (Martingale property)**.** *Each coordinate $\{\pi_{g,k}\}_{g \geq 0}$ is a bounded $\mathcal{F}_g$-martingale: $\mathbb{E}[\pi_{g+1,k} \mid \mathcal{F}_g] = \pi_{g,k}$.*

Lemma 2 (Vertex absorption)**.** *The simplex vertices $\{e_1, \dots, e_K\}$ are the only absorbing states of the chain $\{\pi_g\}$. With probability one, there exists a random index $J \in \{1, \dots, K\}$ such that $\pi_g \to e_J$, with absorption probabilities*

$$\mathbb{P}(J = k) \;=\; \pi_{0,k}, \qquad k = 1, \dots, K.$$

The expected first-extinction time is finite for every $K \geq 2$ and every $n \geq 1$.

Lemma 3 (Monotone extinction)**.** *The extinct set is non-decreasing: $E_g \subseteq E_{g+1}$ almost surely. Equivalently, once $\pi_{g,k} = 0$ for some g, then $\pi_{g',k} = 0$ for all $g' \geq g$.*

Proof sketch. Lemma 1 follows from $\mathbb{E}[N_{g,k} \mid \mathcal{F}_g] = n\pi_{g,k}$ for the multinomial counts in (**??**). Lemma 2 is a standard application of the bounded martingale convergence theorem combined with the Feller-style argument for finite Markov chains with absorbing states; the absorption probabilities follow from $\mathbb{E}[\pi_{\infty,k}] = \pi_{0,k}$ and $\pi_{\infty,k} \in \{0, 1\}$ almost surely. Lemma 3 is immediate: if $\pi_{g,k} = 0$, then $N_{g,k} \mid \mathcal{F}_g \sim \mathrm{Binomial}(n, 0)$ is identically zero, so $\pi_{g+1,k} = 0$. Full proofs are given in Appendix **??**. □

These three lemmas are not new. Lemma 1 appears in essentially this form in Borkar [2025] and Jangjoo et al. [2025], the latter as a consequence of a more general martingale property for sufficient statistics of exponential families under closed-loop learning. Lemma 2 is established for the $K = 2$ case by Suresh et al. [2024] and for the categorical / Random Energy Model case by Jangjoo et al. [2025]; Benati et al. [2025] prove a closely related first-extinction law via Wright–Fisher diffusion. Riis [2026] observes that the same dynamics describe recursive n-gram language model retraining. Lemma 3 is implicit in all of these but, to our knowledge, is not stated as a separate result; we record it explicitly because it is what drives the singularity statement of the next subsection.

Remark 2. Vertex absorption of the mixture weights does *not* by itself imply collapse of the model in any geometric sense relevant to neural training. The component parameters η_k for surviving components continue to be estimated consistently from their respective sub-samples, and the Fisher information of the surviving sub-model is non-singular by Assumption 1. The geometric phenomenon we identify — progressive degeneracy of the full-model Fisher information along the trajectory — is a consequence of *which directions* in Θ become unidentifiable as components go extinct, not of the absorption itself.

D Algebraic proof of Aoyagi–Nagata strict monotonicity

We prove Lemma **??**(2): for every $K \geq 2$ and every $2 \leq r \leq K$,

$$\lambda(K,r) - \lambda(K,r-1) \;=\; 1 - \frac{1}{2(i(K,r)+1)} \;\geq\; \frac{3}{4},$$

where $\lambda(K,r)$ is given by (**??**) and $i(K,r) := \max\{j \in \mathbb{Z}_{\geq 0} : j^2 + j \leq 2(K-r+1)\}$.

Reparametrization. For $K \geq 2$ fixed, set $m := K - r + 1$, so m ranges over $\{1, 2, \ldots, K\}$ as r ranges over $\{K, K-1, \ldots, 1\}$. Define

$$\mu(m) \;:=\; \lambda(K, K-m+1) \;=\; (K-m) + \frac{i(m)^2 + i(m) + 2m}{4(i(m)+1)}, \qquad i(m) \;:=\; \max\{j \in \mathbb{Z}_{\geq 0} : j^2 + j \leq 2m\}.$$

Strict increase of $\lambda(K,\cdot)$ in r is equivalent to strict decrease of $\mu(\cdot)$ in m. We prove $\mu(m) - \mu(m+1) = 1 - 1/(2(i(m)+1))$ for every $m \in \{1, \ldots, K-1\}$, which establishes the closed-form drop and (since $i(m) \geq 1$ for all $m \geq 1$) the lower bound $\geq 3/4$.

Step 1: The Vandermonde index increments by at most one. We show $i(m+1) \in \{i(m), i(m)+1\}$. Set $i := i(m)$. Suppose for contradiction that $i(m+1) \geq i+2$. By definition of $i(m+1) \geq i+2$,

$$(i+2)^2 + (i+2) \;\leq\; 2(m+1), \qquad \text{i.e.,} \qquad 2(m+1) \;\geq\; i^2 + 5i + 6.$$

By definition of $i(m) = i$, the value $j = i+1$ fails the threshold at m:

$$(i+1)^2 + (i+1) \;>\; 2m, \qquad \text{i.e.,} \qquad 2m \;<\; i^2 + 3i + 2.$$

Hence $2(m+1) < i^2 + 3i + 4$. Combined with $2(m+1) \geq i^2 + 5i + 6$, this requires $i^2 + 5i + 6 < i^2 + 3i + 4$, i.e., $2i < -2$, i.e., $i < -1$. This contradicts $i \geq 0$. Hence $i(m+1) \in \{i, i+1\}$.

Step 2: Drop computation, Case A: $i(m+1) = i(m)$. Set $i := i(m) = i(m+1)$. Direct computation:

$$\begin{aligned}
\mu(m) - \mu(m+1) \;&=\; 1 + \frac{i^2 + i + 2m}{4(i+1)} - \frac{i^2 + i + 2(m+1)}{4(i+1)} \\
&=\; 1 + \frac{(i^2 + i + 2m) - (i^2 + i + 2m + 2)}{4(i+1)} \\
&=\; 1 - \frac{2}{4(i+1)} \;=\; 1 - \frac{1}{2(i+1)}.
\end{aligned}$$

Step 3: Drop computation, Case B: $i(m+1) = i(m)+1$. Set $i := i(m)$ and $i' := i(m+1) = i+1$. The transition occurs precisely when $j = i+1$ first satisfies the threshold, i.e., $(i+1)^2 + (i+1) \leq 2(m+1)$ but $(i+1)^2 + (i+1) > 2m$. Since $2m$ and $2(m+1)$ differ by exactly 2, and $(i+1)^2 + (i+1) = (i+1)(i+2)$ is even, the inequality $(i+1)(i+2) \leq 2(m+1)$ combined with $(i+1)(i+2) > 2m = 2(m+1) - 2$ forces

$$(i+1)(i+2) \;=\; 2(m+1), \qquad \text{equivalently} \qquad 2m \;=\; (i+1)(i+2) - 2 \;=\; i^2 + 3i.$$

Substitute $2m = i^2 + 3i$ in the first quotient of $\mu(m)$:

$$\frac{i^2 + i + 2m}{4(i+1)} \;=\; \frac{i^2 + i + i^2 + 3i}{4(i+1)} \;=\; \frac{2i^2 + 4i}{4(i+1)} \;=\; \frac{2i(i+2)}{4(i+1)} \;=\; \frac{i(i+2)}{2(i+1)}.$$

Substitute $2(m+1) = (i+1)(i+2)$ and $i' = i+1$ in the second quotient of $\mu(m+1)$:

$$\frac{(i')^2 + i' + 2(m+1)}{4(i'+1)} \;=\; \frac{(i+1)(i+2) + (i+1)(i+2)}{4(i+2)} \;=\; \frac{2(i+1)(i+2)}{4(i+2)} \;=\; \frac{i+1}{2}.$$

The difference of these two quotients is

$$\frac{i(i+2)}{2(i+1)} - \frac{i+1}{2} \;=\; \frac{i(i+2) - (i+1)^2}{2(i+1)} \;=\; \frac{i^2 + 2i - i^2 - 2i - 1}{2(i+1)} \;=\; -\frac{1}{2(i+1)}.$$

Therefore $\mu(m) - \mu(m+1) = 1 - 1/(2(i+1))$, identical to Case A.

Step 4: The lower bound. For $m \geq 1$, $i(m) \geq 1$ since $j = 1$ satisfies $j^2 + j = 2 \leq 2m$. Hence $i(m) + 1 \geq 2$ and

$$\mu(m) - \mu(m+1) \;=\; 1 - \frac{1}{2(i(m)+1)} \;\geq\; 1 - \frac{1}{4} \;=\; \frac{3}{4}.$$

Translating back via $r = K - m + 1$ gives the claimed bound on $\lambda(K, r) - \lambda(K, r-1)$. □

E Additional theory details

E.1 VaDE latent-prior Fisher bound

We prove the latent-prior Fisher scaling used in Section 2.3. For VaDE, write

$$p_\Theta(x) = \sum_{c=1}^{K} \pi_c q_c(x), \qquad q_c(x) = \int p_\psi(x \mid z) \mathcal{N}(z \mid \mu_c, \Sigma_c)\, dz,$$

and define

$$\rho_c(x) = \frac{\pi_c q_c(x)}{p_\Theta(x)}.$$

For a latent-prior parameter block $a_c \in \{\mu_c, \Sigma_c\}$,

$$\nabla_{a_c} \log p_\Theta(x) = \rho_c(x) \nabla_{a_c} \log q_c(x).$$

Therefore

$$I_{a_c a_c}(\Theta) = \mathbb{E}_{x \sim p_\Theta}\left[\rho_c(x)^2 \nabla_{a_c} \log q_c(x) \nabla_{a_c} \log q_c(x)^\top\right].$$

Taking traces and using $0 \leq \rho_c(x) \leq 1$,

$$\operatorname{tr} I_{a_c a_c}(\Theta) \leq \mathbb{E}_{x \sim p_\Theta}\left[\rho_c(x) \|\nabla_{a_c} \log q_c(x)\|^2\right].$$

Since $\rho_c(x) p_\Theta(x) = \pi_c q_c(x)$, if

$$\mathbb{E}_{x \sim q_c}\left[\|\nabla_{a_c} \log q_c(x)\|^2\right] \leq C_c,$$

then

$$\operatorname{tr} I_{a_c a_c}(\Theta) \leq \pi_c C_c.$$

Thus the component-specific Fisher curvature is $O(\pi_c)$.

If $E = \{c : \pi_c = 0\}$, then $p_\Theta(x)$ no longer depends on the corresponding (μ_c, Σ_c) blocks. Hence the associated score coordinates vanish identically, and

$$\dim \ker I(\Theta) \geq |E| \left(d + \frac{d(d+1)}{2}\right).$$

E.2 Proof of Proposition 1

Let $m_\alpha = \lceil \alpha n \rceil$ be the number of real samples injected at each generation. For any generation g and component k, let $N_{g,k}^{\text{real}}$ be the number of injected real samples assigned to component k. Then

$$N_{g,k}^{\text{real}} \sim \text{Binomial}(m_\alpha, \pi_k^*).$$

Since the total mixed count satisfies

$$N_{g,k}^{(\alpha)} \geq N_{g,k}^{\text{real}},$$

we have

$$\mathbb{P}\Big(N_{g,k}^{(\alpha)} = 0\Big) \leq \mathbb{P}\big(N_{g,k}^{\text{real}} = 0\big) = (1 - \pi_k^*)^{m_\alpha} \leq \exp(-m_\alpha \pi_k^*) \leq \exp(-m_\alpha \pi_{\min}^*).$$

A union bound over K components and G generations gives

$$\mathbb{P}\Big(\exists g < G, \exists k : N_{g,k}^{(\alpha)} = 0\Big) \leq KG \exp(-m_\alpha \pi_{\min}^*) \leq KG \exp(-\alpha n \pi_{\min}^*).$$

Thus, if

$$\alpha \geq \frac{\log(KG/\delta)}{n \pi_{\min}^*},$$

the failure probability is at most δ. On the complementary event, every component receives at least one mixed training sample at every generation, so no component enters the extinct set over the horizon. Hence $r_g = K$ and

$$\lambda_g = \lambda(K, K) = \frac{2K - 1}{2}, \qquad 0 \leq g \leq G.$$

E.3 Final-layer decoder curvature

Let the final decoder layer be $f_W(z) = W\phi(z)$, and consider

$$\mathcal{L}_g(W) = \frac{1}{2\sigma^2}\mathbb{E}_{(z,x)\sim Q_g}\|W\phi(z) - x\|_2^2.$$

Vectorizing W, direct differentiation gives

$$\nabla^2_{\mathrm{vec}(W)}\mathcal{L}_g(W) = \frac{1}{\sigma^2}\mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top] \otimes I_{\mathrm{out}}.$$

Therefore

$$\mathrm{rank}\left(\nabla^2_{\mathrm{vec}(W)}\mathcal{L}_g(W)\right) = \mathrm{rank}\left(\mathbb{E}_{Q_g}[\phi(z)\phi(z)^\top]\right) \cdot \dim(I_{\mathrm{out}}).$$

Thus final-layer decoder curvature degenerates when recursive training concentrates the latent feature distribution.

E.4 Tier 1 — Gaussian mixtures: exact validation of the theory

Question: Does the labeled-mixture theorem happen exactly?

We run the labeled univariate Gaussian mixture recursion of Section C for $K \in \{3, 5, 8\}$ candidate components and sample sizes $n \in \{50, 100, 200, 500, 1000\}$, with $G = 200$ generations for trajectory measurements (Panels A, B, C, E) and $G_{\max} = 30{,}000$ for relaxation-time measurements (Panel D). The data-generating distribution at $g = 0$ has uniform weights $\pi^*_k = 1/K$ and well-separated means $\eta^*_k = 2k$ for $k = 1, \ldots, K$. We use 10 seeds per configuration; figures show mean and 1σ bands. The relaxation time τ is recorded as the first generation at which $r_g = 1$.

Trajectory and closed-form drops. Figure 6 reports the central result. Panel A shows mixture-weight trajectories $\{\pi_{g,k}\}_{k=1}^K$ for $K = 5$, $n = 100$: components drift, hit zero at random generations, and remain extinct, reproducing the multinomial absorption dynamics of Lemmas 1–3. Panel B shows the trajectory RLCT λ_g for $K \in \{3, 5, 8\}$, with horizontal reference lines at the regular case $\lambda(K, K)$ and the singleton-collapsed value $\lambda(K, 1)$ from Lemma **??**. Across all three K values, λ_g is strictly non-increasing and converges toward $\lambda(K, 1)$ as predicted by Theorem **??**(i)–(iii). At $G = 200$, the fraction of seeds reaching $r_g = 1$ is 0.70 for $K = 3$, 0.50 for $K = 5$, and 0.50 for $K = 8$; remaining seeds are still descending the trajectory, consistent with the predicted relaxation-time scale (Panel D). The closed-form drops of Lemma **??**(2) — $3/4$ at extinctions $r = K \to K - 1$ for all K, then $5/6$ and $7/8$ at later transitions for $K \geq 5$ and $K \geq 8$ respectively — are visible as discrete steps in Panel B's trajectories.

Fisher nullity tracks $|E_g|$. Panel C reports the Fisher nullity $|E_g| = K - r_g$ as a function of generation, which by Theorem **??**(iv) lower-bounds $\dim\ker I(\theta_g)$. The empirical mean nullity grows monotonically from 0 at $g = 0$ to approximately 1.4, 3.4, and 6.4 at $g = 200$ for $K = 3, 5, 8$, approaching the asymptotic value $K - 1$ predicted by Lemma 2. The monotone-non-decreasing structure of Panel C is a direct consequence of Lemma 3 and is preserved across all seeds.

Relaxation time scales linearly in n. Panel D reports $\mathbb{E}[\tau]$ as a function of sample size n for $K \in \{2, 3, 5\}$, computed with $G_{\max} = 30{,}000$ to ensure full fixation in all 10 seeds across all configurations (no truncation bias). The empirical relaxation times are linear in n across all K, validating the qualitative scaling of Proposition **??**. The empirical slopes $\mathbb{E}[\tau]/n$ from least-squares fits through the origin are 1.01, 1.18, and 1.59 for $K \in \{2, 3, 5\}$, compared to the asymptotic prediction $2\log K = 1.39$, 2.20, and 3.22. The empirical-to-predicted ratio is 0.73 for $K = 2$, 0.54 for $K = 3$, and 0.49 for $K = 5$. The constant gap is consistent with the $o(n)$ correction in Proposition **??**, which becomes more pronounced for multi-allelic diffusions because the asymptotic constant $2H(\pi^*)$ is approached only as $n \to \infty$. We emphasize that linear-in-n scaling — the qualitative content of Proposition **??** — holds robustly across all K and n tested.

Real-data injection arrests collapse. Panel E reports the active component count trajectory r_g for $\alpha \in \{0, 0.1, 0.25, 0.5\}$ at $K = 5$, $n = 100$. Synthetic-only recursion ($\alpha = 0$) decays from $r_g = 5$ to $r_g \approx 1.6$ over $G = 200$ generations, consistent with the relaxation-time scaling of Panel D. Injection of $\alpha = 0.1$ real data substantially slows the decay, while $\alpha \in \{0.25, 0.5\}$ prevents extinction entirely

over the $G = 200$ horizon: r_g remains pinned at $K = 5$. The transition between drift-dominated ($\alpha = 0$) and mutation-dominated ($\alpha \geq 0.25$) regimes is consistent with the Wright–Fisher-with-mutation phase structure underlying Proposition **??**.

Summary. Tier 1 confirms the qualitative content of Theorem **??** and Proposition **??**: monotone non-increasing λ_g converging to $\lambda(K, 1)$, monotone-non-decreasing Fisher nullity tracking $|E_g|$, linear-in-n relaxation time, and arrested collapse under real-data injection. Quantitative agreement with the asymptotic relaxation constant $2H(\pi^*)$ is partial at the sample sizes tested, with finite-n corrections of order 25–50% for $K \geq 3$, consistent with the $o(n)$ statement of Proposition **??**. The remaining tiers test whether these singular-geometric signatures persist as the model class moves from the labeled mixture to neural generative models.

F Tier 1 — Gaussian mixtures: exact validation of the theory

Question: *Does the labeled-mixture theorem hold exactly in its intended setting?*

We run the labeled univariate Gaussian mixture recursion for $K \in \{3, 5, 8\}$, sample sizes $n \in \{50, 100, 200, 500, 1000\}$, and 10 seeds. Trajectory panels use $G = 200$ generations; relaxation-time estimates use $G_{\max} = 30{,}000$ to avoid truncation bias. The initial mixture has uniform weights $\pi_k^* = 1/K$ and separated means $\eta_k^* = 2k$, and τ denotes the first generation at which $r_g = 1$.

Figure 6 validates the exact mixture mechanism. Panel A shows absorbing multinomial dynamics: components drift, hit zero, and remain extinct. Panel B shows that the trajectory RLCT λ_g is stepwise non-increasing and moves from the regular value $\lambda(K, K)$ toward the singleton-collapsed value $\lambda(K, 1)$, with the closed-form drops of Lemma **??** visible at extinction events: $3/4$ for the first extinction, then $5/6$ and $7/8$ at later transitions for larger K. By generation $G = 200$, the fraction of seeds reaching $r_g = 1$ is 0.70, 0.50, and 0.50 for $K = 3, 5, 8$, respectively; the remaining seeds are still descending the same predicted trajectory. Panel C confirms the Fisher-nullity prediction: the mean extinct-component count $|E_g| = K - r_g$ grows monotonically from 0 to approximately 1.4, 3.4, and 6.4 at $g = 200$ for $K = 3, 5, 8$, approaching the asymptotic value $K - 1$.

Panel D tests the relaxation-time scaling. Across $K \in \{2, 3, 5\}$, $\mathbb{E}[\tau]$ grows linearly in n. Least-squares slopes $\mathbb{E}[\tau]/n$ are 1.01, 1.18, and 1.59, compared with the diffusion-limit constants $2 \log K = 1.39, 2.20, 3.22$. Thus the qualitative linear-in-n prediction holds robustly, while the asymptotic constants show finite-n corrections of roughly 25–50%, consistent with the $o(n)$ term. Panel E shows that real-data injection arrests collapse: for $K = 5, n = 100$, synthetic-only recursion decays to $r_g \approx 1.6$ by $G = 200$, $\alpha = 0.1$ substantially slows the decay, and $\alpha \in \{0.25, 0.5\}$ prevents extinction over the full horizon, keeping $r_g = 5$. Tier 1 therefore validates the exact RLCT and Fisher-nullity mechanism before moving to neural generative models.

We also use the Gaussian-mixture tier to calibrate the SGLD-based LLC estimator used in the neural experiments. In this setting the exact trajectory RLCT is known: $\lambda_g = \lambda(K, r_g)$. Figure 7 shows that the empirical estimates $\widehat{\lambda}_g$ track the closed-form RLCT trajectory across recursive generations, including the stepwise drops induced by component extinction. The calibration achieves correlation 0.96 and mean absolute error 0.29 between $\widehat{\lambda}_g$ and λ_g, supporting the use of LLC decay as a singular-complexity diagnostic in the VaDE and VAE experiments.

G Experimental details and reproducibility

This section gives the experimental details needed to reproduce the results in Section 3. All reported curves are averages over independent random seeds. Unless otherwise stated, shaded regions and error bars denote one standard deviation across seeds.

G.1 Datasets and recursive protocol

For the Gaussian-mixture tier, data are generated synthetically from the specified labeled Gaussian mixture. For the image-model tiers, VaDE and the standard VAE are trained on MNIST, and the DDPM tier is trained on FashionMNIST. Images are resized or kept at 28×28, normalized to $[0, 1]$, and flattened for MLP-based VaDE/VAE models. The DDPM uses the 28×28 image format directly.

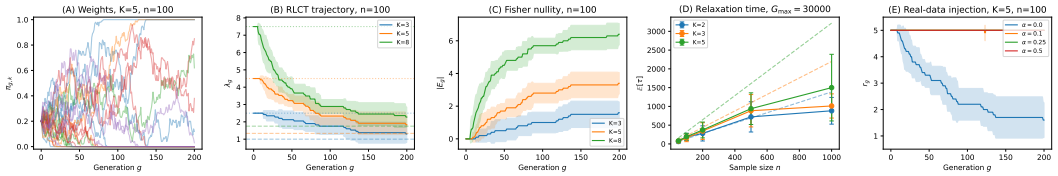


Figure 6: **Theorem 1 validated in labeled univariate Gaussian mixtures.** (A) Mixture-weight trajectories for $K = 5$, $n = 100$: components drift to extinction and remain absorbed. (B) Trajectory RLCT λ_g converges from $\lambda(K, K)$ (dotted) toward $\lambda(K, 1)$ (dashed) for $K \in \{3, 5, 8\}$, with closed-form drops at component-extinction events (Lemma **??**). (C) Fisher nullity $|E_g|$ grows monotonically toward $K - 1$ (Theorem 1(iv)). (D) Relaxation time $\mathbb{E}[\tau]$ is linear in n for $K \in \{2, 3, 5\}$, with empirical slope below the asymptotic-in-n prediction $2 \log K$ at moderate n, consistent with finite-n corrections. (E) Real-data injection ($\alpha > 0$) arrests collapse: $\alpha \geq 0.25$ prevents extinction over $G = 200$. All panels: mean over 10 seeds, 1σ bands.

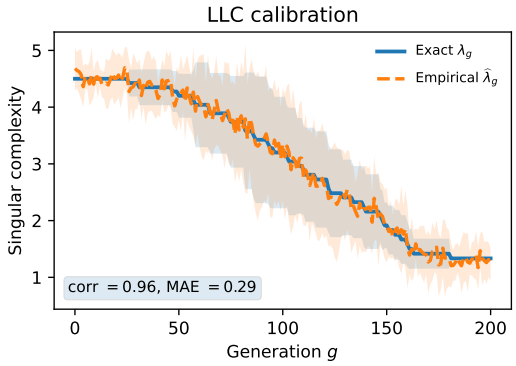


Figure 7: **Calibration of the empirical LLC estimator in labeled Gaussian mixtures.** Empirical LLC estimates $\widehat{\lambda}_g$ track the exact RLCT trajectory $\lambda_g = \lambda(K, r_g)$, with correlation 0.96 and mean absolute error 0.29. This calibrates the SGLD-based LLC estimator used in the neural VaDE and VAE experiments.

Across tiers, recursive training follows

$$\mathcal{D}_{g+1} \sim p_{\theta_g}^{\otimes n}, \qquad \theta_{g+1} = \mathrm{Train}(\mathcal{D}_{g+1}, \theta_g).$$

The α-mixed control replaces an α-fraction of each generation's synthetic samples with fresh samples from the original training distribution:

$$\mathcal{D}_{g+1}^{(\alpha)} = \mathcal{D}_{g+1}^{\mathrm{syn}} \cup \mathcal{D}_{g+1}^{\mathrm{real}}, \qquad |\mathcal{D}_{g+1}^{\mathrm{real}}| = \lceil \alpha n \rceil.$$

We use $\alpha = 0$ for synthetic-only recursion. VaDE and VAE experiments use $\alpha \in \{0, 0.1, 0.25, 0.5\}$, while the DDPM experiment uses $\alpha \in \{0, 0.1, 0.25\}$.

G.2 Tier 1: Gaussian mixtures

For the labeled Gaussian-mixture tier, we use a K-component univariate Gaussian mixture with known unit variance. The initial mixture has uniform weights

$$\pi_k^* = 1/K$$

and separated means

$$\eta_k^* = 2k, \qquad k = 1, \ldots, K.$$

The labeled recursion uses the closed-form MLE update described in Section 2.1. We run

$$K \in \{3, 5, 8\}, \qquad n \in \{50, 100, 200, 500, 1000\},$$

with 10 random seeds. Trajectory plots use $G = 200$ recursive generations. Relaxation-time experiments use $G_{\max} = 30{,}000$ to reduce truncation bias. The relaxation time τ is the first generation at which $r_g = 1$. The exact trajectory RLCT is computed as

$$\lambda_g = \lambda(K, r_g),$$

using the Aoyagi–Nagata formula in Section 2.2.

G.3 Tier 2: VaDE

For VaDE, we use $K = 10$ latent mixture components and latent dimension $d = 10$. The encoder and decoder are multilayer perceptrons. The encoder outputs the variational Gaussian parameters $(\mu_\phi(x), \log \sigma^2_\phi(x))$, and the decoder maps $z \in \mathbb{R}^{10}$ back to the image space. The latent prior is

$$p(z, c) = \pi_c \mathcal{N}(z \mid \mu_c, \Sigma_c).$$

We train on MNIST with

$$n = 5000, \qquad G = 50, \qquad 10 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25, 0.5\}.$$

Each generation is trained by ELBO maximization initialized from the previous generation's weights unless otherwise specified. Components with

$$\pi_{g,k} < 10^{-3}$$

are treated as near-extinct, giving

$$E^\varepsilon_g = \{k : \pi_{g,k} < 10^{-3}\}, \qquad r^\varepsilon_g = K - |E^\varepsilon_g|.$$

For each component k, we compute the normalized latent-prior Fisher curvature

$$\kappa_{g,k} := \frac{1}{d + d(d+1)/2} \operatorname{tr} I_{(\mu_{g,k}, \Sigma_{g,k}), (\mu_{g,k}, \Sigma_{g,k})}(\Theta_g).$$

The predicted near-null count is

$$|E^\varepsilon_g| \left(d + \frac{d(d+1)}{2} \right),$$

which equals $65|E^\varepsilon_g|$ for $d = 10$.

G.4 Tier 3: standard VAE

The standard VAE uses the same MLP encoder/decoder architecture and latent dimension $d = 10$ as the VaDE tier, but replaces the Gaussian-mixture latent prior with a single standard Gaussian prior. We train on MNIST with

$$n = 5000, \qquad G = 50, \qquad 10 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25, 0.5\}.$$

Each recursive generation uses 50 epochs of continuation training. Because the standard VAE has no explicit mixture weights, we evaluate collapse using class-coverage entropy, empirical LLC, Hessian/Gauss–Newton effective ranks, and decoder feature-covariance effective rank rather than latent-component extinction.

G.5 Tier 4: DDPM

The DDPM tier uses a compact U-Net denoising model trained on FashionMNIST. We use

$$G = 20, \qquad 5 \text{ seeds}, \qquad \alpha \in \{0, 0.1, 0.25\}.$$

Each generation trains the denoising model on the generated image set, with the α-mixed control replacing an α-fraction of generated samples with fresh FashionMNIST samples. Distributional collapse is measured by class-coverage entropy and FID. Geometry is measured by the effective rank of U-Net feature covariances and a blockwise empirical Gauss–Newton effective rank computed on the denoising loss.

G.6 Diagnostics

For image models, class-coverage entropy is computed as

$$H_{\mathrm{cov}}(g) = -\sum_{c=1}^{10} q_{g,c} \log q_{g,c},$$

where $q_{g,c}$ is the fraction of generated samples classified as class c by a fixed classifier trained on the corresponding real dataset. For MNIST experiments, the classifier is trained on real MNIST; for FashionMNIST experiments, it is trained on real FashionMNIST.

For a positive semidefinite matrix with eigenvalues $\{\lambda_i\}_{i=1}^m$, we compute effective rank using the entropy definition

$$r_{\mathrm{eff}} = \exp\left(-\sum_{i=1}^{m} \tilde{\lambda}_i \log \tilde{\lambda}_i\right), \qquad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}.$$

When reporting normalized effective rank, we divide by the corresponding initial or maximum reference rank so that values lie on a comparable scale across diagnostics.

Hessian and Gauss–Newton ranks are computed on minibatches from the generation-g training or evaluation set. For VaDE and VAE, decoder feature-covariance rank is computed from the latent features $\phi(z)$ entering the final decoder layer:

$$\widehat{\Sigma}_{\phi,g} = \frac{1}{m}\sum_{i=1}^{m} \phi(z_i)\phi(z_i)^\top.$$

For the DDPM, U-Net feature rank is computed analogously from intermediate U-Net activations, and the blockwise Gauss–Newton rank is computed from selected parameter blocks of the denoising objective.

G.7 Empirical LLC estimation

Empirical local learning coefficient estimates are computed using the SGLD-based estimator of Furman and Lau [Furman and Lau, 2024]. For each checkpoint, we run SGLD chains initialized near the trained parameter θ_g, sample parameter perturbations from the local posterior, and estimate the local learning coefficient from the temperature-scaled increase in empirical loss. The Gaussian-mixture tier calibrates this estimator against the exact trajectory RLCT $\lambda_g = \lambda(K, r_g)$. Robustness checks repeat the LLC computation with doubled chain length and step-size perturbations, as shown in Figure 5D.

G.8 Early-warning prediction

For VaDE, the early-warning target at generation g is

$$\mathcal{C}_{g,\Delta} = \mathbf{1}\{r^{\varepsilon}_{g+\Delta} < r^{\varepsilon}_{g}\}, \qquad \Delta = 10.$$

We train logistic predictors using three feature groups: distribution-only features, geometry-only features, and their union. Distribution-only features include active component count, class-coverage entropy, and held-out negative log-likelihood. Geometry-only features include componentwise latent-prior Fisher curvature, Fisher near-nullity, empirical LLC, Hessian effective rank, Gauss–Newton effective rank, and decoder feature-covariance rank. Prediction is evaluated using leave-one-seed-out splits, so checkpoints from the same recursive trajectory never appear in both the training and test folds. We report mean AUROC and one standard deviation across held-out-seed folds.

G.9 Statistical reporting

All reported means and uncertainty bands are computed across independent random seeds. Unless otherwise stated, curves and tables report mean $\pm$ one standard deviation. The Gaussian-mixture, VaDE, and VAE experiments use 10 seeds. The DDPM experiment uses 5 seeds because each recursive diffusion run is more computationally expensive. No normality assumption is used to interpret the bands as confidence intervals.

G.10 Compute resources

The Gaussian-mixture experiments are lightweight and can be run on CPU. The VaDE and VAE experiments can be run on a single GPU or CPU, although GPU training is faster. The DDPM tier is the most expensive experiment and is run on GPU. Approximate runtimes depend on hardware and implementation details; in our runs, the Gaussian-mixture tier required minutes on CPU, VaDE/VAE recursive runs required hours on a single GPU, and the DDPM recursive runs required substantially longer because a denoising model is trained at each generation.

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